

- 9.1 (a) Show how two 2-to-1 multiplexers (with no added gates) could be connected to form a 3-to-1 MUX. Input selection should be as follows:
- If  $AB = 00$ , select  $I_0$
  - If  $AB = 01$ , select  $I_1$
  - If  $AB = 1 - (B \text{ is a don't-care})$ , select  $I_2$

- (b) Show how two 4-to-1 and one 2-to-1 multiplexers could be connected to form an 8-to-1 MUX with three control inputs.

- (c) Show how four 2-to-1 and one 4-to-1 multiplexers could be connected to form an 8-to-1 MUX with three control inputs.

- 9.2** Design a circuit which will either subtract  $X$  from  $Y$  or  $Y$  from  $X$ , depending on the value of  $A$ . If  $A = 1$ , the output should be  $X - Y$ , and if  $A = 0$ , the output should be  $Y - X$ . Use a 4-bit subtracter and two 4-bit 2-to-1 multiplexers (with bus inputs and outputs as in Figure 9-7).

**9.4** Realize a full adder using a 3-to-8 line decoder (as in Figure 9-17) and  
(a) two OR gates.

(b) two NOR gates.

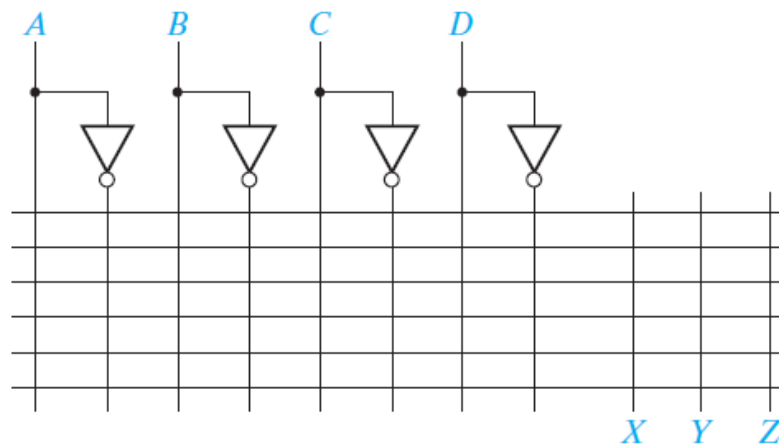
**9.8** The following PLA will be used to implement the following equations:

$$X = AB'D + A'C' + BC + C'D'$$

$$Y = A'C' + AC + C'D'$$

$$Z = CD + A'C' + AB'D$$

- (a) Indicate the connections that will be made to program the PLA to implement these equations.



- (b) Specify the truth table for a ROM which realizes these same equations.

**9.9** Show how to implement a full subtracter using a PAL. See Figure 9-33.

**9.10** (a) If the ROM in the hexadecimal to ASCII code converter of Figure 9-26 is replaced with a PAL, give the internal connection diagram.

(b) If the same ROM is replaced with a PLA, give the PLA table.

**9.13** Expand the following function about the variable  $b$ .

$$F = ab'cde' + bc'd'e + a'cd'e + ac'de'$$

**9.15** Show how to make a 4-to-1 MUX, using an 8-to-1 MUX.

- 9.17** 2-to-1 multiplexers with an active high output and active high enable are to be used in the following implementations:
- (a) Show how to implement a 4-to-1 multiplexer with an active high output and no enable using two of the 2-to-1 MUXes and a minimum number of additional gates.

(b) Repeat part (a) for a 4-to-1 multiplexer with an active low output.

- (c) Repeat part (b) assuming the output of the 2-to-1 MUX is 1 (rather than 0) when the enable is 0.

- 9.20** Realize the function  $f(a, b, c, d, e) = \Sigma m(6, 7, 9, 11, 12, 13, 16, 17, 18, 20, 21, 23, 25, 28)$  using a 16-to-1 MUX with control inputs  $b, c, d$ , and  $e$ . Each data input should be 0, 1,  $a$ , or  $a'$ . (*Hint:* Start with a minterm expansion of  $F$  and combine minterms to eliminate  $a$  and  $a'$  where possible.)